[INCH-POUND]
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SUPERCEDING
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April 20, 1987

COMMERCIAL ITEM DESCRIPTION

DISC, ABRASIVE (CLOTH & FIBRE)

The General Services Administration has authorized the use of this commercial item description.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This commercial item description covers industrial quality abrasive coated cloth and fibre discs used in portable and fixed plate sanding machines for polishing, finishing, grinding and removal of rust and scale.

1.2 Classification. The abrasive discs shall be of the following types, grits, backings, coatings, grades, styles, classes, bonds and sizes.

Type I - Aluminum oxide, fused brown, fused white, or calcined white unless otherwise specified.

Type II - Alumina zirconia alloy.

Type III - Silicon Carbide.

Type IV - Ceramic aluminum oxide.

Type V - Garnet.

Grit size - 24, 36, 40, 50, 60, 80, 100, 120, 150, 180, 220(66.0), 240(53.5), 280(44.0), 320(36.0), 400(23.6), 600(16.0) as specified. Equivalent micron size shown in parenthesis for fine grits.

Cloth backing - "J" weight (jeans cloth), "X" weight (drills cloth), "Y" weight (heavy drills or sateen cloth), or "F" (Fibre) as specified.

Coating C - Closed coat (100% coverage of the backing by the abrasive grain).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Engineering Group (7FXEE), 819 Taylor Street, Fort Worth, TX 76102

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Coating O - Open coat (40% to 70% coverage of the backing by the abrasive grain).

Grade W. Waterproof for wet or dry applications.

Grade D. Nonwaterproof for dry applications.

Grade N. Nonwaterproof and surface treated with zinc stearate to reduce loading.

Style A - Plain flat, no arbor hole, no adhesive backing.

Style B - Plain flat, with arbor hole, no adhesive backing.

Class 1 - 1/2 inch arbor hole (nonstandard).

Class 2 - 5/8 inch arbor hole.

Class 3 - 7/8 inch arbor hole.

Style C - Plain flat, no arbor hole, with pressure-sensitive adhesive (PSA) coating for attachment, with protective liner, or other means of protecting adhesive backing. Including discs furnished on continuous roll form without liner.

Style D - Plain flat, with quick change mechanical locking device for attachment.

Class 1 - Turn-on style with female thread on disc back.

Class 2 - Twist-on style with male thread on disc back.

Class 3 - Snap-on style with locking plastic cam design.

Class 4 - "Velcro" type backing.

Bond A - Glue over glue.

Bond B - Resin over glue.

Bond C - Resin over resin.

Bond D - Waterproof (resin).

Disc size - 1, 1-1/2, 2, 3, 4, 4-1/2, 5, 6, 7, 8, 9, 9-1/8, 10, 11, 12, (outside diameter in inches) as specified.

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2. APPLICABLE DOCUMENTS

2.1 The documents referenced in this commercial item description shall be the issues in effect on the date of the invitation for bids or request for proposal unless otherwise specified. These documents form a part of this commercial item description to the extent specified. In the event that there is a conflict between this commercial item description and a document referenced herein, this commercial item description shall take precedence.

Federal Standard:

FED-STD-376 - Preferred Metric Units For General Use By The Federal Government.

Copies of Federal specifications and standards are available from the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Copies of military specifications and standards may be obtained from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

ASTM standard:

D828 - Test Methods for Tensile Breaking Strength of Paper and Paperboard.
D1682 - Standard Test Methods for BREAKING LOAD AND ELONGATION OF TEXTILE FABRICS.
E 29 - Standard Practice For Using Digits In Test Data To Determine Conformance With Specifications.

Application for copies of ASTM standards should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.

ANSI standards:

B7.7 - Safety Requirements For Abrading Materials With Coated Abrasives Systems.
B74.18 - Specifications For Grading Of Certain Abrasive Grain On Coated Abrasive Products.

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Application for copies of ANSI standards should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.

3. SALIENT CHARACTERISTICS

3.2 Abrasive grain. Particle size distribution shall conform with the limits specified in ANSI B74.18. The abrasive grain shall be 95% free from extraneous material.

3.3 Adhesive. The making and sizing coats shall be glue or a comparable adhesive, or synthetic resin as specified. When waterproof abrasive discs are specified, the adhesive shall be a good quality resin which will minimize dislodging of the grain from the backing under hard wet or dry usage of the coated product. Bonding of the abrasive grain to the backing shall be equal to or better than that considered good commercial quality.

3.4 Cloth backing material. The treated cloth backing material shall be in accordance with Table 1 and tested in accordance with paragraph 4.7.1. When grade W discs are specified, the backing material shall be treated to make it impervious to water.

3.4.1 Fibre backing material. The fibre backing material shall be vulcanized fibre or a functionally similar synthetic and shall be in accordance with Table 2 and tested in accordance with paragraph 4.7.2. The backing shall have the strength and flexibility to grind surfaces at high speeds. The backing material shall be not less than 0.023 inch thick for grit sizes 80 and finer and not less than 0.028 inch thick for grit sizes 60 and coarser.

Table 1

Minimum average treated cloth breaking strength (lbs/in)		
Cloth type	Warp direction	Filling direction
Jeans ("J")	80	35
Drills ("X")	130	50
Heavy Drills ("Y")	160	65

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Table 2

Minimum average fibre breaking strength (lbs/in)	
Machine direction	Transverse direction
350	180

3.5 Disc attachment design.

3.5.1 Style A. Self explanatory, (see 1.2).

3.5.2 Style B. Self explanatory, (see 1.2).

3.5.3 Style C. The back of the disc shall be coated with a pressure-sensitive adhesive (PSA) capable of retaining the disc on a flat surface flexible holder designed for self-adhesive discs during operation. The PSA shall be protected with a peel-off liner or other means of protecting the adhesive backing.

3.5.4 Style D. The back of the disc shall incorporate a quick-change type of fastening device which permits rapid application and removal of the disc on the holder without the use of tools. The device shall retain the disc on the holder and provide safe operation at full recommended disc speed.

3.5.4.1 Class 1. This device shall incorporate a metal flange with female threading compatible for use on a Norton Company "Speed-Lok" disc holder.

3.5.4.2 Class 2. This device shall incorporate a plastic/nylon metal flange with male threading compatible for use on a 3M Company "Roloc" disc holder.

3.5.4.3 Class 3. This device shall incorporate raised plastic/nylon cams designed to operate safely with a Merit Abrasive Products Company "POWER-LOCK" disc holder.

3.5.4.4 Class 4. This device shall incorporate a backing of loop "Velcro" type backing suitable for attachment to a disc holder having hook pile "Velcro" type material.

3.6 Bonds. Bond shall be A, B, C, or D as specified. Bonds shall be as defined in ANSI B74.18.

3.7 Dimensional tolerance. The disc outside diameter and arbor hole diameter as specified shall be accurate to within ± 1/16 inch.

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3.8 Slits. When slits are specified, discs shall have radial incisions with no appreciable width, equidistantly spaced.

3.9 Slots. When slots are specified, discs shall have radial arranged slots 1/16 to 1/8 inch wide.

3.10 Workmanship. The abrasive discs shall have evenly dispersed grain. The making and sizing coats of bonding agents shall adequately bind the grain to the backing during intended use. Discs shall be suitable for use under the guidelines described in ANSI B7.7.

3.11 Identification marking. Coated abrasive discs shall be marked in a permanent and legible manner with the abrasive grit size.

3.12 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description unless otherwise specified.

3.13 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED. STD. No. 376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

3.14 Commercial item. The use of the term "commercial item description" in this document does not imply that any item or items offered are not required to conform with all requirements specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The levels of inspection shall be in accordance with MIL-STD-105.

4.3 Lot formation. In the absence of manufacturer's lot identification, an inspection lot shall consist of all abrasive discs of the same type, grade, class, style and size and produced by the same manufacturer offered for delivery at one time shall be considered a lot for purposes of inspection. For test purposes, a lot may not consist of more than one identifiable manufacturer's lot.

4.4 Sampling for visual and dimensional examination. A random sample of abrasive discs shall be selected from each lot in accordance with inspection level S-2, AQL 6.5 expressed in terms of percent defective. The sample unit shall be one disc.

4.5 Sampling for tests. A random sample of treated cloth or fibre backing material shall be selected from each lot. The number of samples selected shall be in accordance with paragraphs 4.7.1, 4.7.2 and 4.7.3 as applicable. Additionally, a sufficient number of samples shall be selected to perform the test specified in 4.7.4. The sample unit for tests specified in 4.7.1, 4.7.2 and 4.7.3 shall be cut strips of the size specified. The sample unit for the test specified in 4.7.4 shall be one disc or "jumbo" production roll if end-item product marking provides adequate traceability to the production roll.

4.6 Visual and dimensional examination. The sample discs shall be visually and dimensionally examined to verify compliance with this description. Tolerance limits specified are absolute limits as defined in ASTM E 29.

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4.7 Testing.

4.7.1 Cloth breaking strength. Breaking strength of the treated cloth, directly after conditioning for at least 24 hours at 7Ø degrees + or - 2 degrees Fahrenheit and 5Ø percent + or - 2% relative humidity shall be determined in accordance with the cutstrip method of ASTM D1682, using 1 inch strips of the coated abrasive product. Values obtained from each specimen in both directions shall be recorded. Conformance shall be determined on the basis of average values computed from all specimens tested.

4.7.2 Fibre breaking strength. Breaking strength of the fibre, directly after conditioning for at least 24 hours at 7Ø degrees + or - 2 degrees Fahrenheit and 5Ø percent + or - 2% relative humidity shall be determined by ASTM D828, except the jaws shall be five inches apart and five specimens each, either one-half inch or one inch wide, shall be taken longitudinally and transversely. Values obtained from each specimen in both directions shall be recorded. Conformance shall be determined on the basis of average values computed from all specimens tested.

4.7.3 Test for waterproofness (Grade W only). 6 samples shall be selected from each inspection lot of waterproof product and subjected to the following procedure: After conditioning each sample by submerging in boiling water for a minimum of 15 minutes, subject each sample to the following test. After sufficient cooling to prevent burning of the skin, the disc shall be rolled around a 1 inch (or smaller) diameter metal rod (after removal of mechanical locking device for Style D), then gently rub each sample between the thumb and forefinger. If the grain coating is dislodged or if the backing paper tears or is observed to have lost it's body, the sample shall have failed testing.

4.7.4 Grading of abrasive grain. The grading, control and tolerance of Types I, III, and V abrasive shall be in accordance with ANSI B74.18. Types II and IV are not covered by ANSI B74.18 and shall be graded in accordance with the manufacturer's standard practice.

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4.8 Certification of test requirements. Unless otherwise specified in the contract or purchase order, a certificate of conformance may be furnished in lieu of actual testing on a lot by lot basis providing that:

1) The product being furnished was produced by a domestic manufacturer.
2) Complete testing is performed at the beginning of the contract period and at least once each year thereafter. Tests shall be performed by the supplier or by a test facility acceptable to the Government.
3) The product being offered has been manufactured by the same manufacturer, using the same material, processes and manufacturing techniques as the product tested.
4) The Government shall have the right to require additional testing (at the suppliers expense) if there is reason to believe that the product or products do not conform with stated requirements.
5) By furnishing a certificate of conformance in lieu of performing tests the supplier agrees to perform or have performed as applicable additional tests at the request of the Government when deemed necessary.

4.9 Examination for preparation for delivery. An examination of unit, intermediate and shipping containers (as applicable) shall be made to determine conformance with the preservation, packaging, packing, labeling and marking required in the contract or order. Samples shall be selected in accordance with inspection level S-2, AQL 6.5 expressed in terms of defects per hundred unit. The sample unit shall be one fully prepared shipping container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

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6. NOTES

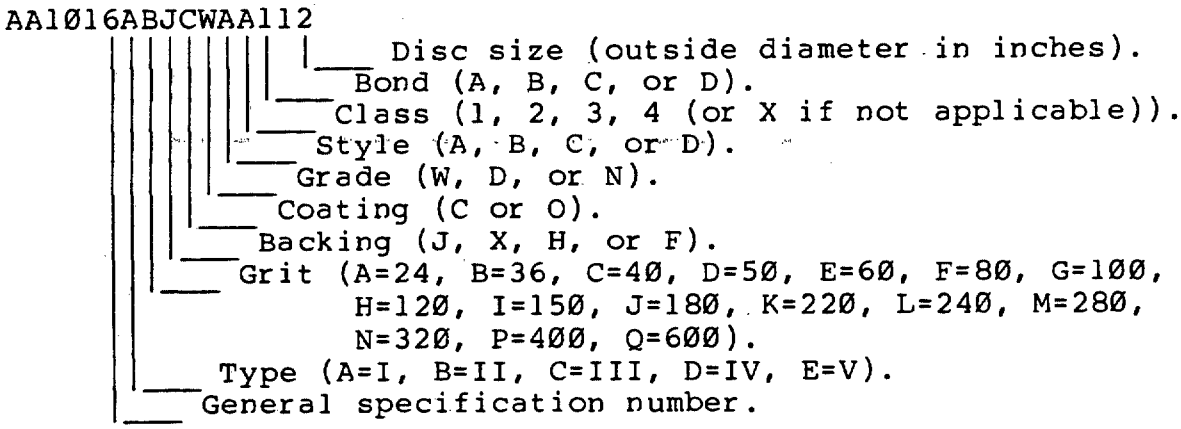
6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents (if applicable).

a. Title, number and date of this commercial item description.
b. Type, grit size, backing material, coating type, grade, style (and class if applicable), bond and disc size, (see 1.2).
c. If slits or slots are required (see 3.8 and 3.9).
d. If certification of testing is not permitted, (see 4.8).
e. The preservation, packaging, packing, labeling, and marking.

6.2 The use of brand names in paragraphs 3.5.4 through 3.5.4.4 are intended only as reference standards to insure product consistency. Reference to these brand names is not an endorsement and discs procured using this commercial item description should be procured only on the basis of the requirements specified herein without regard for brand.

6.3 Users of coated abrasive discs are encouraged to consult and observe the safety and use guidelines contained in ANSI B7.7 (see 2.1)

6.4 Part Numbering. Abrasive discs conforming with this commercial item description shall be identified by a part number configuration consisting of identification of a portion of the specification number, type, grit, backing weight, coating, grade, style, class, bond and size. An example of the part number configuration is shown below. This part numbering system is intended for identification and cross-indexing of the item within the Federal cataloging system. Part numbers are not required to be placed on the product or container.



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MILITARY INTERESTS:

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Review activity:

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PREPARING ACTIVITY:

GSA - FSS.

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